

deep self-attention network. In particular, instead of computing the hidden states from scratch for each new segment, we reuse the hidden states obtained in previous segments. The reused hidden states serve as memory for the current segment, which builds up a recurrent connection between the segments. As a result, modeling very long-term dependency becomes possible because information can be propagated through the recurrent connections. Meanwhile, passing information from the previous segment can also resolve the problem of context fragmentation. More importantly, we show the necessity of using relative positional encodings rather than absolute ones, in order to enable state reuse without causing temporal confusion. Hence, as an additional technical contribution, we introduce a simple but more effective relative positional encoding formulation that generalizes to attention lengths longer than the one observed during training.

Transformer-XL obtained strong results on five datasets, varying from word-level to character-level language modeling. Transformer-XL is also able to generate relatively coherent long text articles with *thousands of* tokens (see Appendix E), trained on only 100M tokens.

Our main technical contributions include introducing the notion of recurrence in a purely self-attentive model and deriving a novel positional encoding scheme. These two techniques form a complete set of solutions, as any one of them alone does not address the issue of fixed-length contexts. Transformer-XL is the first self-attention model that achieves substantially better results than RNNs on both character-level and word-level language modeling.

2 Related Work

In the last few years, the field of language modeling has witnessed many significant advances, including but not limited to devising novel architectures to better encode the context (Bengio et al., 2003; Mikolov et al., 2010; Merity et al., 2016; Al-Rfou et al., 2018), improving regularization and optimization algorithms (Gal and Ghahramani, 2016), speeding up the Softmax computation (Grave et al., 2016a), and enriching the output distribution family (Yang et al., 2017).

To capture the long-range context in language modeling, a line of work directly feeds a representation of the wider context into the network

as an additional input. Existing works range from ones where context representations are manually defined (Mikolov and Zweig, 2012; Ji et al., 2015; Wang and Cho, 2015) to others that rely on document-level topics learned from data (Dieng et al., 2016; Wang et al., 2017).

More broadly, in generic sequence modeling, how to capture long-term dependency has been a long-standing research problem. From this perspective, since the ubiquitous adaption of LSTM, many efforts have been spent on relieving the vanishing gradient problem, including better initialization (Le et al., 2015), additional loss signal (Trinh et al., 2018), augmented memory structure (Ke et al., 2018) and others that modify the internal architecture of RNNs to ease the optimization (Wu et al., 2016; Li et al., 2018). Different from them, our work is based on the Transformer architecture and shows that language modeling as a real-world task benefits from the ability to learn longer-term dependency.

3 Model

Given a corpus of tokens $\mathbf{x} = (x_1, \dots, x_T)$, the task of language modeling is to estimate the joint probability $P(\mathbf{x})$, which is often auto-regressively factorized as $P(\mathbf{x}) = \prod_t P(x_t | \mathbf{x}_{<t})$. With the factorization, the problem reduces to estimating each conditional factor. In this work, we stick to the standard neural approach to modeling the conditional probability. Specifically, a trainable neural network is used to encode the context $\mathbf{x}_{<t}$ into a fixed size hidden state, which is multiplied with the word embeddings to obtain the logits. The logits are then fed into the Softmax function, yielding a categorical probability distribution over the next token.

3.1 Vanilla Transformer Language Models

In order to apply Transformer or self-attention to language modeling, the central problem is how to train a Transformer to effectively encode an arbitrarily long context into a fixed size representation. Given infinite memory and computation, a simple solution would be to process the entire context sequence using an unconditional Transformer decoder, similar to a feed-forward neural network. However, this is usually infeasible with the limited resource in practice.

One feasible but crude approximation is to split the entire corpus into shorter segments of man-